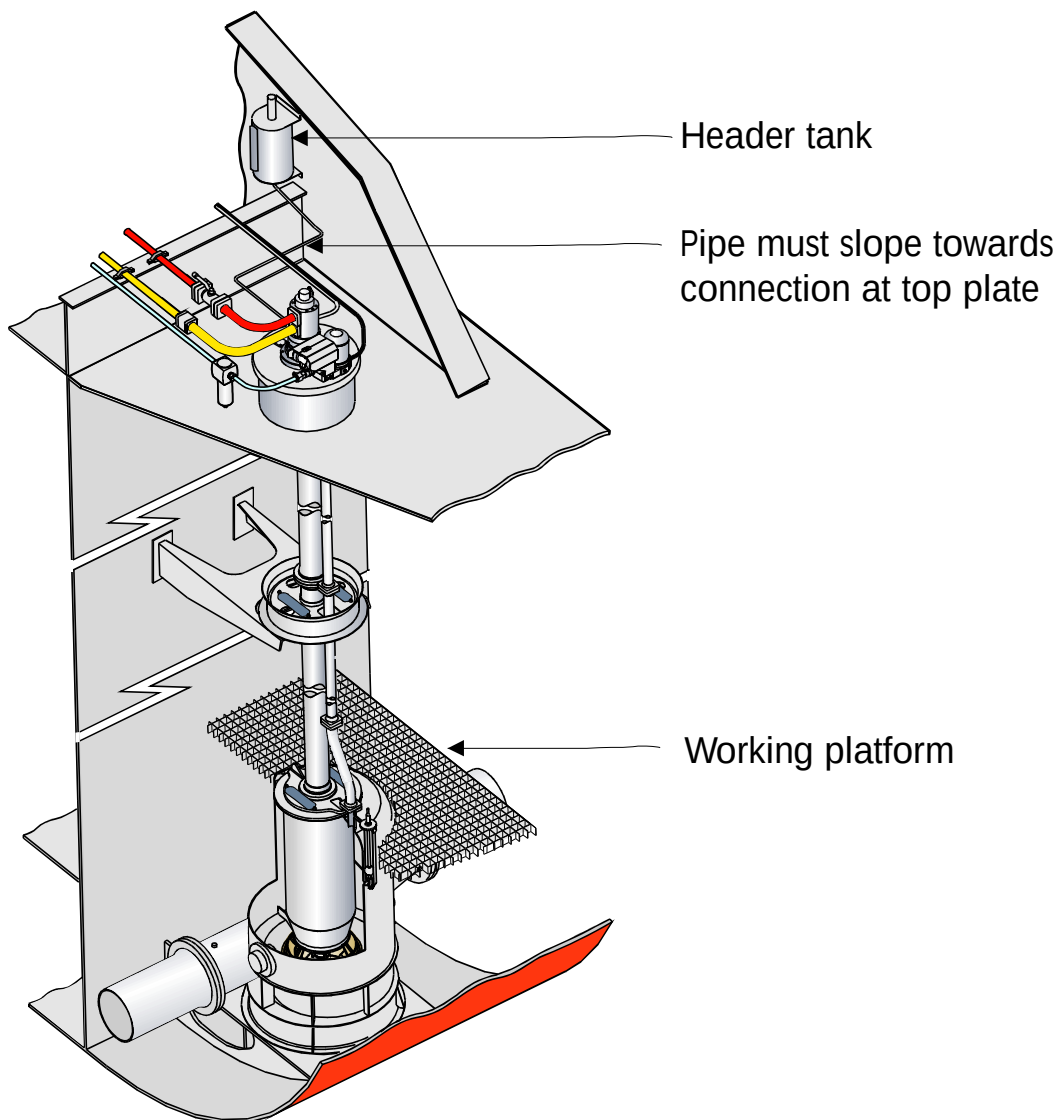


Service Bulletin

SB-Ballast Pumps - Liquid in Cofferdam





Experience has shown that good quality and correct type of cofferdam liquid is important in order to get a long lifetime of the seal arrangement.

In case of change in cofferdam level, the level should be monitored over a certain period to ensure the problem is identified before starting doing service of the pump.



No. 1000-0134-4

LOG FORM FOR LIQUID FILLED COFFERDAM ON FRAMO SUBMERGED BALLAST PUMPS

VESSEL NAME: _____ TYPE OF LIQUID USED IN COFFERDAM: _____

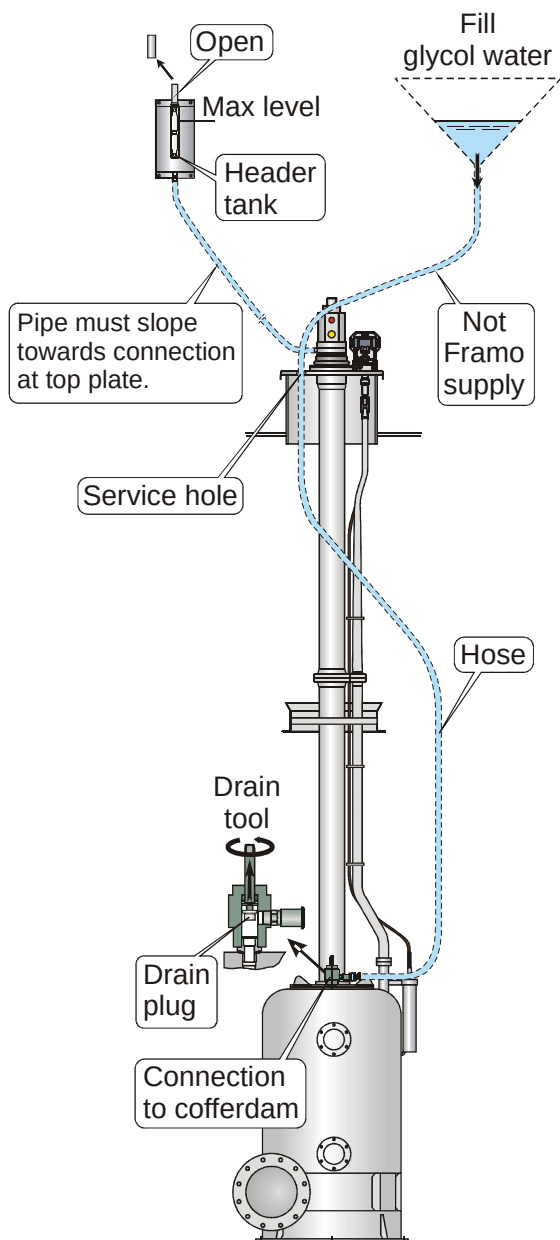
Header tank level to be measured in mm above low level mark on sight glass. 10 mm on sight glass is equivalent to:

- For SB200: 0,12 litres volume.
- For SB300/400/600: 0,36 litres volume.

[illegible]

The procedure 0164-0062-4 in this Service Bulletin specify the correct cofferdam liquid, the volume in each pump and also the method to get a safe filling of the cofferdam.

All the SB-pumps shall have liquid filled cofferdam.



Cofferdam liquid:

- Demineralised water: 40 - 50 %
- Glycol * 60 - 50 %
- Freezing point -40°C - -35°C

* Propylene glycol, C₃H₈O₂
(monopropylene glycol, 1.2 propanediol, propanediol)
or ethylene glycol, C₂H₆O₂
(monoethylene glycol, 1.2 ethanediol, ethanediol)

Use pure glycol without additives. Pure glycol is clear and colourless.

Note!

Think safety and health. Read the Material Safety Data Sheet before using the different glycol.

Note!

Never use other anti-freeze solutions. Some of the additives in anti-freeze forms deposits on seal parts and thereby causing seals to fail.

Filling:

- Hydraulic oil should be filled into pump before filling of cofferdam.
- Install the drain tool and unscrew the drain plug. Connect the hose to the drain tool, and then fill the cofferdam through the hose while venting at the header tank.
- Fill the cofferdam until liquid level stabilises at max. level mark on the sight glass. Ensure to keep up a continuous flow of liquid to avoid air pockets in the cofferdam filling hose.

Topping up cofferdam:

If required , top up the header tank until liquid level is at "normal" level, ref. operation manual.

The cofferdam volume depends on pump type and pump length. The following formulas give approx. volume (in litres) required to fill cofferdam to high level mark in header tank:

Pump type	Pump head	Pipe stack	Header tank	Total
SB200	2,5	L x 0,5	2,5	5 + (L x 0,5)
SB300	80	L x 3,5	12	92 + (L x 3,5)
SB400	95	L x 3,5	12	107 + (L x 3,5)
SB600	125	L x 3,5	12	137 + (L x 3,5)

L = Pump length in metres. (See Framo Specification)



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